CS917 Foundations of Computing - Maths and Stats Assignment

This assignment is due **at noon on Thursday 18th December, 2025**. The submission is on Tabula, and should comprise a PDF containing answers. The work that you submit should be **your own work** and please show **full working** where appropriate, as this is necessary to gain full marks. Marks for each question are indicated. The total marks you can get is 100. If you have any questions, please post them on Moodle.

1 Discrete Mathematics

1. For each of the following formulae, find a logically equivalent formula in which $\wedge$, $\Rightarrow$ and $\Leftrightarrow$ do not occur:

   (i) $\neg(p \Rightarrow q)$ [4 marks];

   (ii) $((p \wedge q) \vee r)$ [4 marks];

   (iii) $\neg((p \wedge q) \Leftrightarrow r)$ [4 marks].

   Use the truth table to show that the proposed solution is indeed equivalent with the original one.

   (iv) Consider the expression transformations you performed in this question. Discuss whether there exists a systematic method (algorithm or general strategy) for converting any propositional formula into a form that uses only $\neg$ and $\vee$. Describe the method informally, and comment on whether such a method always terminates and why [3 marks].

   [total 15 marks]

2. Write out paraphrases of the following, using $\forall$, $\exists$ and $=$

   (i) Frodo has a ring [2 marks];

   (ii) Sauron does not have any rings [2 marks];

   (iii) The One Ring rules all the other rings [2 marks];

   (iv) The ring that Frodo has is the One Ring [2 marks];

   (v) whoever wears the ring, becomes invisible (i.e., no other ordinary human can see that person) [3 marks];

   (vi) Bombadil Tom can see the ring-wearer, hence he is no ordinary human (use the previous statement from (v) to formulate this one) [3 marks].

   Auxiliary clauses: has(x, y): x has/is in possession of y; rules(x, y): x rules y; wears(x, y): x wears y; sees(x, y): x can see y; and ordinary(x): x is an ordinary human. [total 14 marks]

3. Consider an arbitrary binary relation $R \subseteq A \times A$ on a non-empty set A.

   (i) Suppose you are told that R is *both* reflexive and antisymmetric, but you do not know whether it is transitive. Construct a concrete example of such a relation on the set $A = \{1, 2, 3\}$ that is **not** transitive. Explain briefly which property fails and why [3 marks].

   (ii) A student claims:

   "If a relation is irreflexive and symmetric, then it must also be intransitive."

   Is this statement true? Give a justification. If it is true, explain why; if it is false, provide a counterexample on a small finite set and explain why the student's reasoning fails [3 marks].

   (iii) Let R be a relation on a set A. Describe a real-world situation (e.g., social networks, tasks with prerequisites, sports competitions, security policies, etc.) where it would be **desirable** for a relation to be a partial order. Explain which properties (reflexive, antisymmetric, transitive) matter most in your scenario and why failing any of them would cause problems. Your answer should be informal but mathematically accurate [3 marks].

   [total 9 marks]

4. Determine which of the following functions are injective and which are surjective (please provide explanations as well):

   (i) $f : \mathbb{Z} \to \mathbb{N}$, where $\forall n \in \mathbb{Z} : f(n) = n^{2024} + 1$ [2 marks];

   (ii) $g : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$, where $\forall (n, k) \in \mathbb{N} \times \mathbb{N} : g(n, k) = 2^n 3^k 5^{n+k}$ [2 marks];

   (iii) $h : \mathbf{P}(\mathbb{N}) \to \mathbf{P}(\mathbb{N})$, where $\forall A \in \mathbf{P}(\mathbb{N}) : h(A) = \mathbb{N} \setminus A$ (recall what $\mathbf{P}(\mathbb{N})$ means) [2 marks];

   (iv) $k : \mathbb{N} \to \mathbb{Z}$, where $\forall n \in \mathbb{N} : k(n) = (-1)^n$ [2 marks].

   (v) Suppose you are designing a function from a domain A to a codomain B. Explain, in general, how the sizes and structures of A and B constrain whether it is possible for a function to be injective, surjective, or bijective. Illustrate your explanation with at least one example not covered in parts (i)–(iv) [4 marks].

   [total 12 marks].

2 Probability and Statistical Analysis

1. There are 3 white and 5 black balls, from which 4 balls are transferred into an empty vessel. From this vessel a ball is drawn and it is found to be white. What is the probability that out of 4 balls transferred, 3 are white and 1 is black? Explain your answer with full working. [10 Marks]

2. The probability of detecting tuberculosis in X-ray examination of a person suffering from the disease is $(1 - b)$. The probability of diagnosing a healthy person as tubercular is a. The ratio of tubercular patients to the whole population is c. Find the probability that a person is healthy if, after the examination, he is diagnosed as tubercular. Explain your answer with full working. [10 Marks]

3. A school wanted to know students' preferences about homework through voting. The students were divided into three groups based on their preferences:

- G_1 is the group of students who preferred more homework.
- G_2 is the group of students who preferred less homework.
- G_3 is the group of students who preferred no change to the homework setup.

30% students belonged to G_1, 50% belonged to G_2 and 20% belonged to G_3. They participated in the voting to decide this matter, with the following student participation.

- 65% of G_1 voted.
- 82% of G_2 voted.
- 50% of G_3 voted.

A student is randomly selected from those who voted. What is the probability that this student is from G_3? Explain your answer with full working. [10 Marks]

4. It is shown that conventional fertilizers can help plants grow by an average of 10 inches. A company claims that their new fertilizer can cause plants to grow more than 10 inches on average. Check this claim using the p-value test with the significance level 0.05. You are given 15 plants that have a sample mean of 11.4 inches and a sample standard deviation of 2.5 inches. Consult the t-distribution table (Fig. 1) as given below for estimating the p-value. Explain your answer with full working.

5. The results below are the calorie content of the different brands of beef and poultry hotdogs.

   - Beef hotdogs: $186, 181, 176, 149, 184, 190, 158, 139, 175, 148, 152, 111$
   - Poultry hotdogs: $129, 132, 102, 106, 94, 102, 87, 99, 170, 113$

   Can it be inferred that the calorie content of poultry hotdogs is lower than the calorie content of beef hotdogs? Use significance level 0.05. Explain your answer with full working. [10 Marks]

	90%	95%	97.5%	99%	99.5%	99.95%	1-Tail Confidence Level
	80%	90%	95%	98%	99%	99.9%	2-Tail Confidence Level
	0.100	0.050	0.025	0.010	0.005	0.0005	1-Tail Alpha
df	0.20	0.10	0.05	0.02	0.01	0.001	2-Tail Alpha
1	3.0777	6.3138	12.7062	31.8205	63.6567	636.6192	
2	1.8856	2.9200	4.3027	6.9646	9.9248	31.5991	
3	1.6377	2.3534	3.1824	4.5407	5.8409	12.9240	
4	1.5332	2.1318	2.7764	3.7469	4.6041	8.6103	
5	1.4759	2.0150	2.5706	3.3649	4.0321	6.8688	
6	1.4398	1.9432	2.4469	3.1427	3.7074	5.9588	
7	1.4149	1.8946	2.3646	2.9980	3.4995	5.4079	
8	1.3968	1.8595	2.3060	2.8965	3.3554	5.0413	
9	1.3830	1.8331	2.2622	2.8214	3.2498	4.7809	
10	1.3722	1.8125	2.2281	2.7638	3.1693	4.5869	
11	1.3634	1.7959	2.2010	2.7181	3.1058	4.4370	
12	1.3562	1.7823	2.1788	2.6810	3.0545	4.3178	
13	1.3502	1.7709	2.1604	2.6503	3.0123	4.2208	
14	1.3450	1.7613	2.1448	2.6245	2.9768	4.1405	
15	1.3406	1.7531	2.1314	2.6025	2.9467	4.0728	
16	1.3368	1.7459	2.1199	2.5835	2.9208	4.0150	
17	1.3334	1.7396	2.1098	2.5669	2.8982	3.9651	
18	1.3304	1.7341	2.1009	2.5524	2.8784	3.9216	
19	1.3277	1.7291	2.0930	2.5395	2.8609	3.8834	
20	1.3253	1.7247	2.0860	2.5280	2.8453	3.8495	
21	1.3232	1.7207	2.0796	2.5176	2.8314	3.8193	
22	1.3212	1.7171	2.0739	2.5083	2.8188	3.7921	
23	1.3195	1.7139	2.0687	2.4999	2.8073	3.7676	
24	1.3178	1.7109	2.0639	2.4922	2.7969	3.7454	
25	1.3163	1.7081	2.0595	2.4851	2.7874	3.7251	
26	1.3150	1.7056	2.0555	2.4786	2.7787	3.7066	
27	1.3137	1.7033	2.0518	2.4727	2.7707	3.6896	
28	1.3125	1.7011	2.0484	2.4671	2.7633	3.6739	
29	1.3114	1.6991	2.0452	2.4620	2.7564	3.6594	
30	1.3104	1.6973	2.0423	2.4573	2.7500	3.6460	

Figure 1: T-distribution table